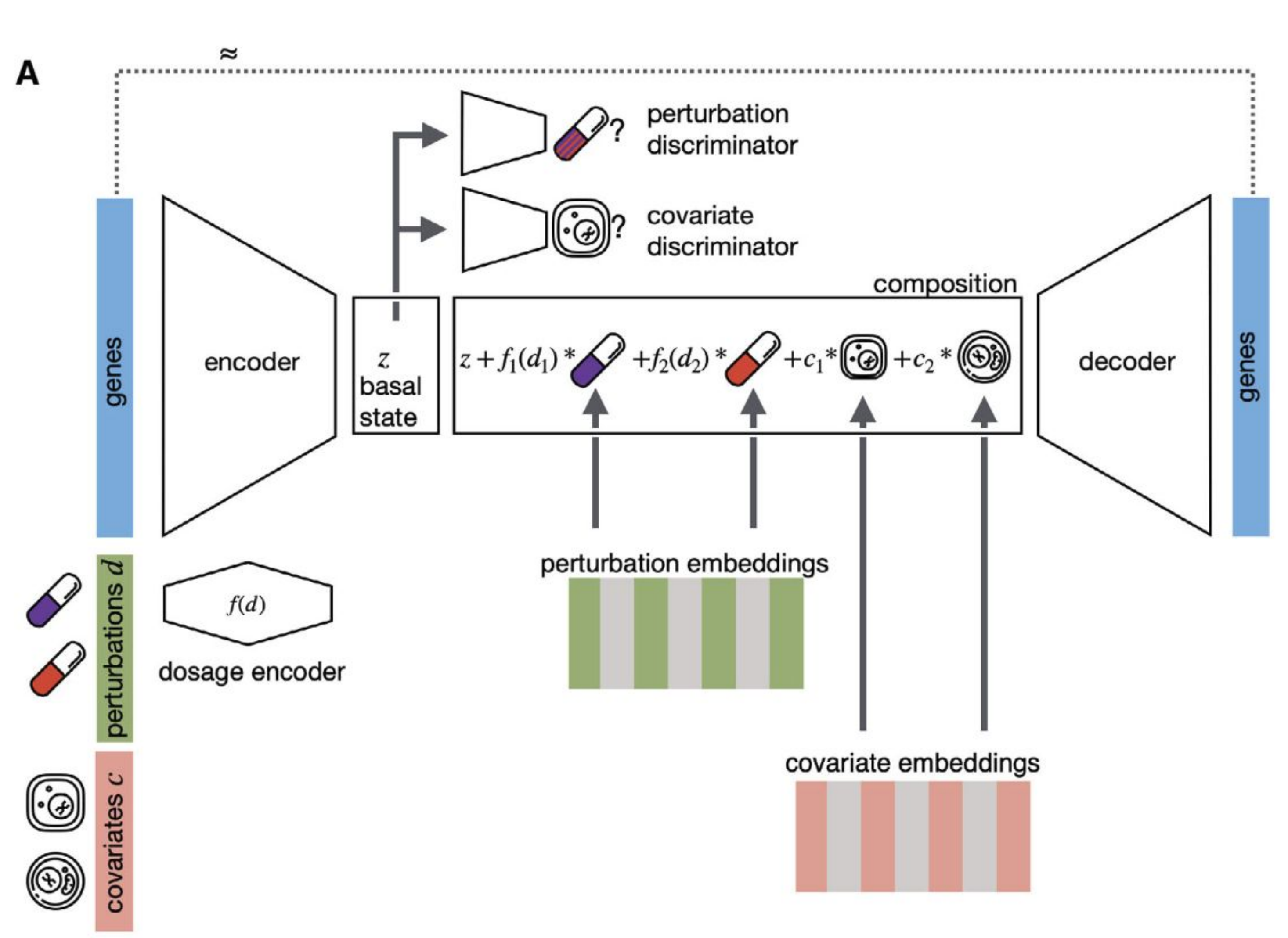
Zefeng Xu, David Naranjo, Moses Yang, Minh Le Tran

# DeepPerturb

Interpretable Cell State Predictions

## Introduction

The ability to predict cellular responses to perturbations like drugs is an indispensable tool for drug development. However, designing software with the flexibility and scalability necessary has proven a long standing issue. Lotfollahi et al. designed the Compositional Perturbation Autoencoder (CPA), an interpretable deep learning architecture based on an autoencoder and adversarial training, to describe and predict single cell responses to perturbations in a scalable way.




## Methodology

### Data

We used a publicly available scRNA dataset from the paper containing the gene expression (how much mRNA exists) of perturbed single cells. The dataset was preprocessed with log-normalization and highly variable genes selected.

| | Gene 1 | Gene 2 |
|---|---|---|
| Cell 1 | 5 | 2 |
| Cell 2 | 1 | 3 |

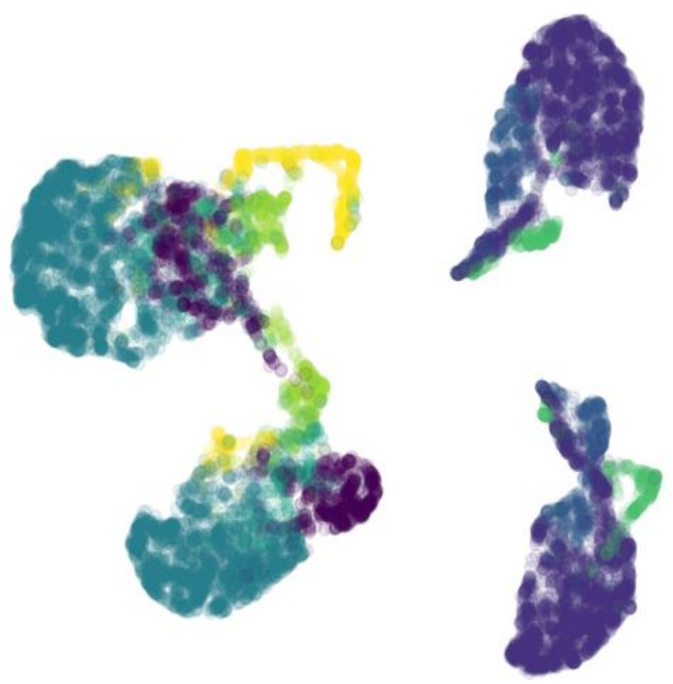
The dataset additionally contains a quantification of the exact conditions the cells were under (metadata). These include the covariates, describing information such as experimental batch and cell type, and the perturbations, detailing direct influences on the cells such as drugs and CRISPR-Cas9 mediated gene knockouts.

### Architecture

CPA is a conditional autoencoder that takes in the cell state, covariates, and perturbations. Through the encoding process, the cell state is embedded as to learn the basal state, separate from the effects of the covariates and perturbations. This separation is learned by simultaneous adversarial training of the encoder against a perturbation discriminator and a covariate discriminator.

The basal cell state embedding is then combined with the perturbation embeddings scaled by doses (scale learned with nonlinear function) and covariate embeddings before being decoded into an approximation of the original cell state (input to encoder). In application, the basal state can be combined with other perturbation and covariate embeddings, yielding a prediction of the cell state under new conditions.

## Results


basal state by cell type


original state by cell type

Right: The original cells are clustered by their cell type.

Left: The basal state is designed to remove any such effects, which results in the cells mixed together without separate clusters.

## Discussion

CPA succeeds in disentangling the basal or unperturbed cell state from an observed perturbed one, allowing for quantitatively evaluating the effects of each possible combination of perturbations like different combination of doses or completely different drugs.

Our model has not been able to both remove perturbation and covariate effects from observed cell states nor reconstruct the input to the extend the author's could. We believe this stems from differences in terms of model implementation such as hyperparameter choices, loss functions, and evaluation metrics than ours. Future work may be dedicated to tuning these or building on CPA to create new models. One option can be to optimize the adversarial training part, which can be highly unstable, through use of a baseline function or similar modification